

Inverse Design of Light-Trapping Layers

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Abstract

This report details the surrogate optimization and active learning approaches used to inversely design light-trapping layers.

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1 Introduction

Introduce the problem of light-trapping layers here [1].

2 Methods and theory

2.1 Rigorous coupled wave analysis

This section aims to introduce the concept of rigorous coupled wave analysis (RCWA), a method in numerical photonics suited well for stacked periodic structures in the x and y axis with a period Λ_x, Λ_y respectively, and where each layer is homogeneous in the z-direction [2]. The structure's optical properties are completely described by the relative permittivity and relative permeability of each layer, meaning that each structure will have a geometry described by

$$\epsilon(x, y, z)_N = \epsilon(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z} \quad (1)$$

and

$$\mu(x, y, z)_N = \mu(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z}, \quad (2)$$

where m and n are integers and the subscript N denotes the N 'th layer of the structure. For the entirety of this project, the designed fourier patterned light trapping layers will be designed using silicon (Si), Titanium Dioxide (TiO_2) and Silicon Nitride (Si_3N_4), which are all nonmagnetic, so the relative permeability $\mu = 1$. The equations actually solved for are of course the Maxwell equations, here given in the form

(3)

(4)

2.2 Machine Learning Surrogates

2.3 Optimization Strategies

3 Results

Present the performance of the surrogate models and the final inverse design outcomes.

Figure 1: Placeholder for a results figure.

4 Discussion

Discuss the results and what worked/didn't work.

5 Outlook

Discuss future work.

6 Conclusion

Summarize the findings.

References

- [1] A. Author and B. Writer. A study on light-trapping nanostructures. *Journal of Nanophotonics*, 10:1–10, 2026.
- [2] Changhyun Kim and Byoungcho Lee. Torcwa: Gpu-accelerated fourier modal method and gradient-based optimization for metasurface design. *Computer Physics Communications*, 282:108552, 2023.